Aim: Different Platform (Python/java), make 3 functions (Factorial, Cube, Square).

1. **Create a New Project**
2. **Create New Web Service**

Code:

```
/*
 * To change this template, choose Tools | Templates
 * and open the template in the editor.
 */
package TYCS;

import javax.jws.WebService;
import javax.jws.WebMethod;
import javax.jws.WebParam;

/**
 *
 * @author Administrator
 */
@WebService(serviceName = "WSPGM")
public class WSPGM {

    /**
     * This is a sample web service operation
     */
    @WebMethod(operationName = "hello")
    public String hello(@WebParam(name = "name") String text) {
        return "hello " + text + "!";
    }

    /**
     * Web service operation
     */
    @WebMethod(operationName = "fact")
```

```
    public String fact(@WebParam(name = "a") int a) {
        int result = 1;
        for (int i = 1; i <= a; i++) {
            result = result * i;
        }
        return "Factorial of number " + a + " is " + result;
    }

    /**
     * Web service operation
     */
    @WebMethod(operationName = "square")
    public float square(@WebParam(name = "n") float n) {
        return n * n;
    }

    /**
     * Web service operation
     */
    @WebMethod(operationName = "cube")
    public float cube(@WebParam(name = "n") float n) {
        float result = (float) java.lang.Math.pow(n, 3);
        return result;
    }
}
```

Output:

WSPGM Web Service Tester

This form will allow you to test your web service implementation (WSDL File)

To invoke an operation, fill the method parameter(s) input boxes and click on the button labeled with the method name.

Methods :

public abstract float tycs.WSPGM.square(float)
square ( )

---

public abstract java.lang.String tycs.WSPGM.fact(int)
fact ( )

---

public abstract java.lang.String tycs.WSPGM.hello(java.lang.String)
hello ( )

---

public abstract float tycs.WSPGM.cube(float)
cube ( )

Square Function Output:

square Method invocation

Method parameter(s)

Type	Value
float	2

Method returned

float : "**4.0**"

Factorial Function Output:

fact Method invocation

Method parameter(s)

Type	Value
int	5

Method returned

java.lang.String : "**Factorial of number 5 is 120**"

Cube Function Output:

cube Method invocation

Method parameter(s)

Type	Value
float	3

Method returned

float : "**27.0**"